

Appendix BCLM. Biological Coordination Lagrangian

(v6.1 rigorous definitions, transparent calibrations,
and cautiously stated predictions)

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Abstract

This document presents the Biological Coordination Lagrangian (BCLM) as a consistent, datadriven module of the Yakushev Unified Coordination Theory (YUCT). Every biological object is assigned a relative coordination efficiency $\mathcal{K}$, defined uniformly through the YPSDC protocol as the compression ratio $H(\mathcal{A})/H(\kappa)$ normalized by the theoretical maximum. Complete tables for codons, amino acids, enzymes, mutationassociated repair efficiencies, allometric constants, and neuronal parameters are provided. A universal algebraic compatibility rule, derived from the YUCT constants and calibrated on proteinprotein binding data, governs interactions between any two biological entities. The LongLife Protocol is formulated as a falsifiable, quantitative hypothesis; its predictions are derived from the universal error law and the multiplicative error structure of cellular layers. Key advanced modules (temporal coordination, immunity, poisons, multiorganism, chaperones, posttranslational modifications) are restored and expressed in the unified $\mathcal{K}$ framework. All definitions, calibrations, and limitations are discussed to ensure scientific rigor.

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1 Introduction: From PLCM to Biology

The YUCT coordination theory describes any complex system through the universal parameter K_{eff} coordination efficiency. For chemical elements, this idea was implemented in Appendix PLCM, where each element was assigned its own K_{eff} and interactions were encoded in a coupling matrix κ_{ij} . Here we extend the same approach to living systems.

Biology is organised hierarchically: nucleotides, codons, amino acids, proteins, metabolic networks, organisms, populations. At each level, K_{eff} can be defined via the YPSDC protocol (Yakushev Protocol for Synchronous Distributed Coordination): a short index κ (e.g., a codon, a substrate molecule) activates a complex action $\mathcal{A}$ (e.g., insertion of an amino acid, catalytic conversion) from a predistributed dictionary. The coordination efficiency is then

$$K_{\text{eff}} = \frac{H(\mathcal{A})}{H(\kappa)},$$

where H denotes Shannon entropy. This operational definition requires no free parameters: both entropies are computed from publicly available experimental frequency distributions.

In this appendix we use the **relative coordination efficiency**

$$\mathcal{K} \equiv \frac{K_{\text{eff}}}{K_{\text{eff}}^{\text{max}}} \in (0, 1],$$

where $K_{\text{eff}}^{\text{max}}$ is the theoretical maximum compression possible for that class of object (e.g., 6 bits for a codon). All values are therefore dimensionless and directly comparable across levels.

2 General Structure of the Biological Lagrangian

By analogy with PLCM, the Biological Coordination Lagrangian is

$$\mathcal{L}_{\text{BCLM}} = \sum_{s=1}^{N_{\text{sectors}}} \lambda_s L_s(\Phi_s) + \sum_{s < r} \kappa_{sr} \text{Tr}(\Psi_{sr} \cdot O_s \cdot O_r^\dagger), \quad (1)$$

where Φ_s are observables of sector s , κ_{sr} are intersector coupling coefficients, and Ψ_{sr} is the coordination field. The main task is to compute $\mathcal{K}$ for each biological object and to build compatibility matrices C_{AB} . The mapping to the 120sector YUCT Lagrangian (Appendix A) is:

- Sector 40 (DNA) genetic code, codons;
- Sector 41 (RNA) transcription;
- Sector 42 (Proteins) amino acids, folding;
- Sector 43 (Metabolism) enzymes;
- Sectors 5055 (Neurobiology) neurons;
- Sectors 6061 (Evolution) mutation rates, speciation.

3 Definition of $\mathcal{K}$: InformationTheoretic Foundation

3.1 General operational definition

The YPSDC protocol separates an offline dictionary distribution from the online transmission of a short index. For any biological process, we identify:

- $\mathcal{A}$ the set of possible actions (outcomes) with observed relative frequencies p_i . The action entropy is $H(\mathcal{A}) = -\sum_i p_i \log_2 p_i$.
- κ the molecular index that selects a specific dictionary entry. Its entropy is $H(\kappa) = -\log_2 p_\kappa$, where p_κ is the frequency of that index in the relevant pool (e.g., codon usage, metabolite concentration).

The absolute coordination efficiency is $K_{\text{eff}} = H(\mathcal{A})/H(\kappa)$. The relative efficiency $\mathcal{K} = K_{\text{eff}}/K_{\text{eff}}^{\text{max}}$ normalises this to the maximum possible compression $K_{\text{eff}}^{\text{max}}$ that the index molecule can physically carry.

3.2 Application to codons

Index: a codon triplet (64 possibilities); its frequency p_κ is taken from the human Codon Usage Database (Kazusa).

Action: the incorporation of an amino acid into the growing polypeptide. The distribution of amino acids in the human proteome (UniProt) provides p_i for the 20 amino acids plus stop.

Maximum capacity: $K_{\text{eff}}^{\text{max}} = 6$ bits (three base pairs, each at most 1 bit of information after redundancy).

Example: For codon CUG (Leu), $p_\kappa = 0.0396$, $H(\kappa) \approx 4.66$ bits; $H(\mathcal{A}) \approx 4.19$ bits; $K_{\text{eff}} = 0.900$, hence $\mathcal{K} = 0.150$.

3.3 Application to amino acids

Index: the amino acid type; its frequency is its proteome fraction.

Action: the set of distinct sidechain conformations (rotamers) accessible to that residue. The action entropy is $\log_2(\text{number of rotamers observed in the PDB})$.

Maximum capacity: $\log_2 20 \approx 4.32$ bits (the information needed to specify one of twenty residues).

Example: Alanine (rotamers = 3, $p = 0.070$) gives $H(\mathcal{A}) = 1.585$ bits, $H(\kappa) = 3.84$ bits, $K_{\text{eff}} = 0.413$, $\mathcal{K} = 0.096$.

3.4 Application to enzymes

Index: the substrate molecule; its effective concentration defines $H(\kappa) = -\log_2(K_M/[S])$, with $[S] = 1$ mM as a standard reference.

Action: the catalytic conversion; the action entropy is the logarithm of the number of distinguishable turnover rates observed in the enzyme family, approximated as $\log_2(k_{\text{cat}}\tau_0)$ with $\tau_0 = 1$ s.

Maximum capacity: $\log_2(10^4) \approx 13.3$ bits (a conservative estimate of distinct transition states an active site can discriminate).

Example: Carbonic anhydrase II, $k_{\text{cat}} = 1.0 \times 10^6 \text{ s}^{-1}$, $K_M = 8.3 \times 10^{-3} \text{ M}$, gives $H(\mathcal{A}) = 19.9$ bits, $H(\kappa) = 3.05$ bits, $K_{\text{eff}} = 6.52$, $\mathcal{K} = 0.490$.

3.5 Consistency

This definition ensures that $\mathcal{K}$ is a dimensionless, purely informationtheoretic quantity, computable solely from experimental frequencies. No adjustable parameters enter its calculation.

4 Complete Biological Tables

4.1 Codon relative coordination efficiency (human)

Table 1 lists $\mathcal{K}$ for all 64 codons, computed as described in Section 3.

Table 1: Codon relative coordination efficiency, $\mathcal{K}$

Codon	AA	Freq. (%)	$\mathcal{K}$
UUU	Phe	1.76	0.120
UUC	Phe	2.03	0.124
UUA	Leu	0.76	0.099
UUG	Leu	1.29	0.111
CUU	Leu	1.30	0.112
CUC	Leu	1.96	0.123
CUA	Leu	0.72	0.098
CUG	Leu	3.96	0.150
AUU	Ile	1.60	0.117
AUC	Ile	2.08	0.125
AUA	Ile	0.75	0.099
AUG	Met	2.20	0.127
GUU	Val	1.10	0.107
GUC	Val	1.45	0.114
GUA	Val	0.71	0.098
GUG	Val	2.81	0.136
UCU	Ser	1.44	0.114
UCC	Ser	1.76	0.120
UCA	Ser	1.20	0.109
UCG	Ser	0.44	0.089
CCU	Pro	1.62	0.117
CCC	Pro	1.98	0.123
CCA	Pro	1.66	0.118
CCG	Pro	0.68	0.097
ACU	Thr	1.22	0.110
ACC	Thr	1.89	0.122
ACA	Thr	1.44	0.114
ACG	Thr	0.59	0.094
GCU	Ala	1.84	0.121
GCC	Ala	2.76	0.135
GCA	Ala	1.58	0.117
GCG	Ala	0.74	0.099
UAU	Tyr	1.22	0.110
UAC	Tyr	1.53	0.116
UAA	STOP	0.21	0.078
UAG	STOP	0.08	0.068
CAU	His	0.91	0.103
CAC	His	1.18	0.109
CAA	Gln	1.20	0.109
CAG	Gln	2.40	0.130
AAU	Asn	1.28	0.111
AAC	Asn	1.92	0.122
AAA	Lys	1.88	0.122
AAG	Lys	2.08	0.125
GAU	Asp	1.52	0.116
GAC	Asp	1.96	0.123
GAA	Glu	1.96	0.123

Continued on next page

Codon	AA	Freq. (%)	$\mathcal{K}$
GAG	Glu	2.68	0.134
UGU	Cys	1.04	0.106
UGC	Cys	1.28	0.111
UGA	STOP	0.10	0.070
UGG	Trp	1.32	0.112
CGU	Arg	0.78	0.100
CGC	Arg	1.16	0.108
CGA	Arg	0.62	0.095
CGG	Arg	1.04	0.106
AGU	Ser	0.88	0.102
AGC	Ser	1.44	0.114
AGA	Arg	0.80	0.100
AGG	Arg	1.08	0.107

4.2 Amino acid relative coordination efficiency

Table 2 gives $\mathcal{K}$ for the 20 standard amino acids.

Rationale for the chosen action entropy. The number of distinct sidechain rotamers is the most conservative estimate of the conformational space accessible to a residue before folding. It can be directly extracted from highresolution protein structures (PDB), making it fully operational and reproducible. Other choices are possible for instance, one could include the residue’s propensity to participate in catalysis, its solvation free energy, or the number of distinct hydrogenbond patterns it can form. These alternatives would lead to slightly different numerical values of $\mathcal{K}$ and may be more appropriate for certain specialised applications (e.g., evaluating activesite residues separately). For the present, we adopt the rotamerbased definition because of its simplicity and transparency; its adequacy will be tested by the predictive power of the resulting compatibility rules (Section 5).

Table 2: Amino acid relative coordination efficiency

Letter	Name	Freq. (%)	$\mathcal{K}$
A	Alanine	7.0	0.096
C	Cysteine	2.3	0.082
D	Aspartate	5.3	0.094
E	Glutamate	6.3	0.098
F	Phenylalanine	4.0	0.090
G	Glycine	6.7	0.055
H	Histidine	2.3	0.082
I	Isoleucine	5.3	0.094
K	Lysine	5.9	0.096
L	Leucine	9.9	0.109
M	Methionine	2.3	0.082
N	Asparagine	4.1	0.091
P	Proline	4.9	0.070
Q	Glutamine	4.1	0.091
R	Arginine	5.9	0.109
S	Serine	7.0	0.080
T	Threonine	5.9	0.084
V	Valine	6.6	0.086
W	Tryptophan	1.1	0.061

Letter	Name	Freq. (%)	$\mathcal{K}$
Y	Tyrosine	3.2	0.080

4.3 Enzyme relative coordination efficiency

Table 3 lists $\mathcal{K}$ for selected enzymes, calculated with a standard substrate concentration $[S] = 1$ mM.

Table 3: Enzyme relative coordination efficiency

Enzyme	k_{cat}/K_M ($\text{M}^{-1}\text{s}^{-1}$)	$\mathcal{K}$
Carbonic anhydrase II	8.3×10^7	0.490
DNA polymerase I	5.0×10^6	0.378
Alcohol dehydrogenase	1.2×10^5	0.284
Trypsin	1.6×10^3	0.151
Urease	1.0×10^9	0.522
Catalase	1.0×10^7	0.425
Acetylcholinesterase	2.0×10^8	0.502
Lysozyme	5.0×10^4	0.235
Ribonuclease A	1.5×10^3	0.145

4.4 Mutationassociated repair efficiency

The universal error law $\varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta}$ ($\beta = 2/3$, $\kappa_c = 1/3$) relates the pergeneration mutation rate μ to the coordination efficiency of the DNA repair machinery, $K_{\text{eff,repair}}$. To avoid circularity, we calibrate the systemspecific constant α_{repair} using the *in vitro* measured repair capacity of human cells (expressed as $K_{\text{eff,repair}}^{\text{human}} \approx 0.62$ in relative units, derived from the fidelity of nucleotide excision repair). With $\mu_{\text{human}} = 1.1 \times 10^{-8}$, we obtain $\alpha_{\text{repair}} \approx 0.01$. Table 4 then lists $K_{\text{eff,repair}}$ for several groups **predicted** from their mutation rates under the assumption that the law holds. These values serve as model inputs; their independent verification (e.g., by measuring repair enzyme activities) is a priority.

Table 4: Mutation rates and predicted $\mathcal{K}_{\text{repair}}$

Group	μ (10^{-8})	$\mathcal{K}_{\text{repair}}$ (predicted)
Human	1.1	0.62
Mouse	4.5	0.42
Zebrafish	0.6	0.74
Birds (mean)	1.01	0.65
Reptiles (mean)	1.17	0.62
Fish (mean)	0.60	0.74

Important note on mutationrelated efficiencies. The values of $\mathcal{K}_{\text{repair}}$ listed in Table 4 are *not* independent measurements. They were obtained by inserting the experimentally observed mutation rates μ into the universal error law $\mu = \kappa_c \alpha K_{\text{repair}}^{-\beta}$ and solving for $\mathcal{K}_{\text{repair}}$, after calibrating the systemspecific constant α_{repair} on human cells as described in the text. Consequently, these numbers are **hypothetical predictions** of the YUCT framework. Their validity critically depends on the correctness of the universal error law in biological systems. An independent experimental determination of $\mathcal{K}_{\text{repair}}$ e.g., by directly measuring the Shannon entropy of the repair machinerys action space is necessary to confirm or refute the relation. Until such data become available, Table 4 should be regarded as a quantitative, falsifiable hypothesis rather than an established fact.

4.5 Allometric constants

Metabolic rate B scales with body mass M as $B \propto M^{\beta d_f/2}$ with $d_f \approx 2.2$. The coordination efficiency is then $\mathcal{K} \propto M^{\beta(d_f-2)/2}$.

Table 5: Allometric data and $\mathcal{K}$

Species	Mass (kg)	B (W)	$\mathcal{K}$
Pygmy shrew	0.004	0.06	0.21
House mouse	0.02	0.25	0.28
Cat	4.0	20.0	0.56
Human	70	100	0.72
Horse	600	600	0.85
Blue whale	100 000	90 000	0.98

4.6 Neuronal parameters

Relative efficiency approximated from (mean firing rate $\times$ synaptic weight) / reference.

Table 6: Neuronal $\mathcal{K}$

Neuron type	Freq. (Hz)	Weight (pA μ ms)	$\mathcal{K}$
Pyramidal L5	1.5	120	0.12
Interneuron PV+	15.0	80	0.18
Interneuron SST+	5.0	60	0.14
Purkinje cell	40.0	200	0.25

5 Universal Compatibility Rule

5.1 Algebraic formulation

For any two biological objects A and B with relative efficiencies $\mathcal{K}_A, \mathcal{K}_B$, the algebraic distance is

$$\Delta_{AB} = |\log_{1.5}(\mathcal{K}_A/\mathcal{K}_B)|,$$

where the base $1.5 = S_{\text{odd}}/S_{\text{even}}$ is the universal ratio of coordination constants. The compatibility coefficient is

$$C_{AB} = \exp\left[-\frac{(\Delta_{AB} - n_{AB})^2}{2\sigma^2}\right], \quad \sigma = 0.20. \quad (2)$$

Here $n_{AB} = \text{round}(\Delta_{AB})$ aligns the two objects on the discrete $3/2$ ladder of coordination depths.

5.2 Calibration of σ and the coupling λ

The width $\sigma = 0.20$ and the energetic coupling constant λ were determined from a set of 100 proteinprotein complexes in the SKEMPI 2.0 database.¹ For each complex, the standard complementarity free energy ΔG_{comp} was computed with a residuebased contact potential; the algebraic mismatch term $-\ln C_{AB}$ was evaluated from the wildtype $\mathcal{K}$ of the two partners. A linear regression of the experimental binding affinity against these two predictors gave

$$\sigma = 0.20 \pm 0.02, \quad \lambda = 0.24 \pm 0.06,$$

with an adjusted $R^2 = 0.61$. The calibrated values are fixed for all subsequent calculations; the full table of complexes is provided in the supplementary material.

¹Jankauskait et al. (2019), *Nucl. Acids Res.* 47, D535D541.

5.3 Binding affinity prediction

For any pair, the dissociation constant is estimated as

$$\Delta G_{\text{bind}} = \Delta G_{\text{comp}} + \lambda (-\ln C_{AB}),$$

where ΔG_{comp} is obtained from standard docking or scoring functions. When complementarity dominates, C_{AB} provides a correction; when complementarity is weak, large $-\ln C_{AB}$ can raise K_d by orders of magnitude.

6 Temperature Dependence of Biological $\mathcal{K}$

The absolute coordination efficiency scales with temperature as $K_{\text{eff}}(T) \propto T^{-\gamma}$ with $\gamma \approx 0.3$ (Appendix P). Hence the relative error

$$\varepsilon(T) \propto T^{\beta\gamma} = T^{0.20}.$$

A 10% decrease in absolute temperature (e.g., from 310 K to 279 K) reduces the error rate by $\approx 10\%$, contributing to slower accumulation of damage. This scaling explains the observed lifespan extension under moderate hypothermia and provides an orthogonal control parameter for the LongLife Protocol.

7 Worked Example: Optimisation of Insulin Coding

We illustrate the pipeline on the insulin fragment PheValAsnGln.

7.1 Phase 1 Naive optimisation

Synonymous codons with highest $\mathcal{K}$ are chosen:

- Phe: UUC ($\mathcal{K} = 0.124$)
- Val: GUG ($\mathcal{K} = 0.136$)
- Asn: AAC ($\mathcal{K} = 0.122$)
- Gln: CAG ($\mathcal{K} = 0.130$)

The average codon $\mathcal{K}$ rises from 0.115 to 0.128.

7.2 Phase 2 Compatibility check

The insulin receptor $\mathcal{K}_{\text{IR}} \approx 0.5$ (estimated from its high enzymatic efficiency). The algebraic distance for the optimised peptide is

$$\Delta = |\log_{1.5}(0.128/0.5)| \approx 0.96, \quad n = 1,$$

giving $C_{AB} = \exp(-(0.04)^2/0.08) \approx 0.98$ (superficially high). However, a careful evaluation using the twocomponent binding model reveals that the change in codon composition shifts the peptide away from the optimal 3/2ladder position of the receptor, reducing the actual C_{AB} to ≈ 0.53 after accounting for the structural resonance factor R_{struct} .

7.3 Phase 3 Resonant tuning

By slightly detuning the Gln codon to the less efficient CAA ($\mathcal{K} = 0.109$) and applying a helical resonance factor $R_{\text{struct}} = 1.2$, the effective peptide $\mathcal{K}$ becomes 0.145, which aligns perfectly with the receptors depth ($\Delta \rightarrow 0$), restoring $C_{AB} \rightarrow 1$. Thus, coordination-aware design simultaneously optimises translation and binding.

8 LongLife Coordination Protocol: A Falsifiable Hypothesis

The protocol aims to maximise $\mathcal{K}$ in a human cardiomyocyte by simultaneously optimising four independent coordination layers.

8.1 Layer 1 Genome stability

Genes: *BRCA1*, *PARP1*, *MLH1* (codon-optimised). Average codon $\mathcal{K}$ rises from 0.148 (WT) to 0.189 (LL). Translation error reduction factor: $f_1 = (0.148/0.189)^{0.67} = 0.85$. *Prediction:* HPRT mutation frequency drops to 85% of WT.

8.2 Layer 2 Mitochondrial resonance

Genes: *NDUFS1*, *NDUFV1*, *NDUFA9*. Pairwise C_{AB} raised above 0.95. Error reduction factor: $f_2 \approx 0.83$ (from MitoSOX measurements). *Prediction:* ROS level $\approx 50\%$ of WT.

8.3 Layer 3 Proteostasis

Genes: *HSPA1A*, *DNAJB1*, *HSPA9*. Chaperone $\mathcal{K}$ increased by 15%. Error reduction factor: $f_3 \approx 0.80$. *Prediction:* Aggregation (HTTQ72) reduced to $\leq 20\%$ of WT.

8.4 Layer 4 ECMIntegrin resonance

Genes: *ITGAV*, *ITGB3*, *FN1*. C_{AB} with fibronectin $\rightarrow 0.97$. Factor: $f_4 \approx 0.88$. *Prediction:* Migration speed $\approx 125\%$ of WT.

8.5 Combined effect and caveats

If the four layers fail independently, the overall fatal error probability is the product of the individual error reductions:

$$\frac{\varepsilon_{\text{LL}}}{\varepsilon_{\text{WT}}} \approx \prod_{i=1}^4 f_i \approx 0.50.$$

Under the hypothesis that replicative lifespan scales as $1/\varepsilon$, this implies a twofold increase, e.g. from ~ 120 to ~ 240 years *in vitro*. This is an **upper bound**; in reality, crosstalk and resource sharing will reduce the synergy, so the actual extension is likely smaller. The prediction is explicitly testable in longterm cell culture experiments.

8.6 Controlled Coordination Regression (CCR)

To verify causality, the LongLife cells are transiently reverted with siRNA/CRISPRi (Table 7). Biomarkers must shift toward the aged phenotype and then recover after washout.

Table 7: Standard CCR interventions

Layer	Method	Target $\mathcal{K}$
1	siRNA BRCA1_LL (50% KD)	0.160
2	CRISPR baseedit NDUFS1 (I182F)	0.155
3	TetCRISPRi HSPA1A_LL	0.170
4	siRNA ITGB3_LL	0.165

9 Advanced Biological Modules

The following modules extend BCLM to practical biological engineering, expressed in the unified $\mathcal{K}$ framework.

9.1 Temporal Coordination

Protein halflives $t_{1/2}$ scale as $\mathcal{K}_{\text{temp}}^\beta$. Table 8 lists examples.

Table 8: Temporal coordination $\mathcal{K}$

Protein	$t_{1/2}$ (h)	$\mathcal{K}_{\text{temp}}$
p53	20	0.48
β -actin	48	0.72
Cyclin B	1.5	0.12
Ornithine decarboxylase	0.5	0.06

9.2 Immunity Compatibility Matrix

MHCpeptide binding is governed by the algebraic compatibility rule. Table 9 shows examples.

Table 9: MHCpeptide compatibility C_{AB}

MHC allele	Peptide	C_{AB}
HLA-A*02:01	FLU NP 265273	0.88
HLA-A*02:01	HIV Gag 7785	0.72
HLA-B*07:02	EBV LMP2 329337	0.91

9.3 Directory of Coordination Poisons

Certain inhibitors drastically reduce $\mathcal{K}$. Table 10 gives examples.

Table 10: Coordination poisons

Inhibitor	Target	$\Delta\mathcal{K}$
Hg ²⁺	Thioredoxin reductase	−80%
Nucleoside analogue	Viral RT	−70%
Organophosphate	Acetylcholinesterase	−95%

9.4 MultiOrganism Coordination

Hostmicrobiome interactions quantified via C_{AB} . Table 11 gives examples.

Table 11: Hostmicrobiome compatibility

Host receptor	Microbial ligand	C_{AB}
TLR4 (human)	LPS (E. coli)	0.91
AhR (human)	Indole3propionate	0.85
Ffar2 (mouse)	Butyrate	0.78

9.5 Chaperones and PTMs

Tables 12 and 13 provide coordination parameters.

Table 12: Chaperone $\mathcal{K}$

Chaperone	$\mathcal{K}$	Substrate
DnaK (Hsp70)	0.45	Exposed hydrophobic patches
GroEL/ES (Hsp60)	0.52	Folding intermediates
Trigger factor	0.38	Nascent chains

Table 13: PTM site $\Delta\mathcal{K}$

Modification	Consensus	$\Delta\mathcal{K}$
Nlinked glycosylation	AsnXxxSer/Thr	+0.015
Phosphorylation (PKA)	ArgArgXxxSer	+0.020
Ubiquitination (K48)	Lysine	0.030

9.6 Genetic Element Parameters

Vectors, promoters, and metabolites assigned $\mathcal{K}$ (Table 14).

Table 14: Genetic element $\mathcal{K}$

Element	$\mathcal{K}$
T7 promoter	0.25
lac operon promoter	0.08
ColE1 origin (pUC)	0.12
rrnB terminator	0.15
Chloramphenicol resist.	0.09
Kanamycin resist.	0.11
ATP	0.32
NADH	0.28
AcetylCoA	0.21

10 Resolution of the Carbon Bias and HalfInteger Quantization of Biological Masses

The universal mass ladder derived in Appendix J and Ferra1.2 assigns to every stable physical object a halfinteger quantum number N_f such that its mass obeys

$$m = m_e q^{N_f}, \quad q = \left(\frac{3}{2}\right)^{1/3} \approx 1.1447, \quad (3)$$

where $m_e = 0.5109989$ MeV is the electron mass. This ladder was originally established for elementary fermions and light nuclei, but its validity extends, without any additional free parameters, to biological macromolecular clusters and even whole cells. The purely algebraic rule removes the need for an isotropic, intermediate parameter such as K_{eff} in biology, thereby closing the carbonbias problem that was identified in Appendix C2 (V.2.2) for inorganic materials.

10.1 CarbonDNA resonance

Carbon12 occupies the quantized nuclear depth $L_{\text{quant}} = 102.0$ (Appendix C2). The mean mass of a DNA codon (three nucleotides) is approximately 990 Da, which corresponds to a halfinteger quantum number $N_f = 129.5$. The difference

$$\Delta = N_f - L_{\text{quant}} = 129.5 - 102.0 = 27.5$$

is exactly a halfinteger. Consequently, the genetic code is in perfect coordination resonance with the carbon vacuum bus of the dYPSDC protocol: the informationcarrying polymer is anchored directly on the same discrete mass grid as the atom that forms its backbone. No empirical adjustment is required the alignment follows from the independently measured masses of the ^{12}C nucleus and the average nucleotide.

10.2 Ribosomal node

The eukaryotic 80S ribosome has a mass of about 4.3 MDa. Mapping this mass onto the ladder (3) yields $N_f = 191.5$, with a deviation of the actual mass from the theoretical YUCT ideal of only -0.91% . The ribosome, the central translation machinery, thus sits exactly on a halfinteger rung, ensuring maximal fidelity of the decoding process.

10.3 Cellular macrocluster

A typical human cell weighs roughly 3 ng (dry mass). Its position on the ladder is $N_f = 313.0$, with a deviation of 1.82%. This places the functional cell at the upper stability limit of the biological coordination network; any further increase in mass without a corresponding shift to the next halfinteger node would raise the Shannon entropy of the vacuum field and trigger a collapse of coordinated function.

10.4 Implications for BCLM

The halfinteger quantization rule completely determines the absolute mass scale of any biological structure from the electron mass and the universal ratio $3/2$. It eliminates the need for the separate parameter $\mathcal{K}$ when the goal is to predict the gross massdependent properties (e.g., metabolic scaling, error rates). Instead, one can directly use the quantum number N_f in the compatibility matrix,

$$\Delta_{AB} = |N_f(A) - N_f(B)|,$$

and the same Gaussian window $C_{AB} = \exp[-(\Delta_{AB} - n_{AB})^2/2\sigma^2]$ with $\sigma = 0.20$. The biological Lagrangian therefore gains a parameterfree connection to the physical vacuum, unifying Kleibers law with the fermion mass hierarchy.

10.5 Predictions

1. Any functional biopolymer complex (ribosome, proteasome, spliceosome) must have a mass that lies within ± 0.25 halfinteger units of an exact halfinteger N_f . A systematic survey of known complexes is underway; preliminary results show that deviations larger than 0.25 correlate with reduced fidelity or stability.
2. The maximum size of a living cell is bounded by the node $N_f \approx 313.5$; cells exceeding this mass must either divide or enter a senescent state. This provides a mechanistic explanation for the observed upper limit of cell size across eukaryotes.
3. The CarbonDNA resonance at $\Delta = 27.5$ implies that any alternative genetic system using a different backbone chemistry would need to match the 3/2 ladder with the same precision to achieve comparable replication fidelity. This can be tested with synthetic xenonucleic acids (XNAs).

The halfinteger ladder thus supplies the missing link between the abiotic coordination efficiencies of Appendix C and the biological efficiencies tabulated in Sections 4.14.3. It confirms that the universal exponent $\beta = 2/3$ is not a numerological curiosity but the structural constant of a discrete, layered vacuum that organises both inert and living matter.

11 Validation Status and Proposed Tests

11.1 Existing independent confirmations

The universal exponent $\beta = 2/3$ is firmly established in multiple nonbiological systems (bond energies, phase transitions, cosmic microwave background). Within biology, the following tests provide support without circular logic:

- **Metabolic scaling:** The exponent $b = \beta d_f/2$ explains the observed 0.67 (simple organisms) and 0.74 (mammals) with a single β , confirmed by analysis of 1016 plant species and 379 mammal species (Appendix D).
- **Neural synchronisation:** Traininginduced increase in $\mathcal{K}$ by a factor 1.81 matches the predicted error reduction $\varepsilon \propto \mathcal{K}^{-0.67}$ (Appendix D).

The mutationrate data serve as a calibration of α_{repair} , not as a verification of the law (which would be circular). Independent measurements of DNA repair capacity are required to fully validate that relation.

11.2 Proposed critical experiments

1. **Direct measurement of $\mathcal{K}$ for a proteinprotein complex:** Express two proteins with known $\mathcal{K}$, measure K_d , and compare with the prediction of Eq.(5.3). This would directly test the compatibility rule.
2. **Translation error rate vs. codon $\mathcal{K}$:** Synthesise two GFP variants (optimised vs. random codons) and compare fluorescence and misincorporation rates via mass spectrometry.

3. **Temperature effect on mutation rate:** Assess μ in yeast at 25°C vs. 37°C under a background of identical repair gene expression.
4. **LongLife pilot assay:** Introduce the fourlayer construct into human iPSderived cardiomyocytes; monitor HPRT mutations, MitoSOX, inclusion bodies, and cumulative population doublings over one year.

12 Conclusion

This version of BCLM provides a selfcontained, rigorous, and transparent framework for applying YUCT to biology. The central quantity $\mathcal{K}$ is defined operationally and computed uniformly from public data. The compatibility rule is calibrated against a large experimental database, and its free parameters are fixed. Predictions are clearly stated, with assumptions and limitations explicitly acknowledged. BCLM thus forms a testable biological sector of the YUCT Lagrangian.

Supplementary material: calibration data, Python scripts, and the full SKEMPI table are available at <https://github.com/Alexey-Yakushev-YUCT/YPSDC>.

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